SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: BharatShield AI – Intelligent Mobile Defense Framework for Android

Version

1.0

Prepared For

National Level Hackathon

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Table of Contents

1. Introduction
2. Problem Statement
3. Proposed Solution
4. System Overview
5. System Architecture
6. Key Innovations
7. Technology Stack
8. Framework and Application Design
9. Database Design
10. Functional Requirements
11. Non-Functional Requirements
12. Interface Requirements
13. Performance Requirements
14. Design Constraints
15. Future Scope
16. Conclusion

1. Introduction

With the rapid growth of mobile applications, smartphones have become essential tools for communication, banking, and personal data storage. However, many mobile users unknowingly install applications that may contain malware, spyware, or unsafe permissions that compromise device security and personal privacy.

BharatShield AI is an intelligent mobile security framework designed to automatically analyze newly installed Android applications and detect potential threats using Artificial Intelligence.

The system provides **real-time malware detection, explainable security analysis, behavioral prediction, and collaborative threat intelligence**, ensuring proactive protection for smartphone users.

2. Problem Statement

The number of Android applications available today has increased significantly, and many users install apps without understanding their potential risks.

Malicious applications may:

- Steal personal information
- Access sensitive permissions
- Monitor user activity
- Send data to external servers

Many existing mobile security applications rely on **manual scanning or signature-based detection**, which often fails to detect newly created malware.

To address this issue, BharatShield AI introduces an automated security framework that scans newly installed applications within **10–15 seconds** and detects suspicious behavior using AI-based analysis.

The system generates a **risk score with over 90% detection accuracy**, helping users make informed decisions before interacting with potentially harmful applications.

3. Proposed Solution

BharatShield AI provides a **real-time AI-based mobile security system** that automatically scans applications immediately after installation.

The system performs:

- APK metadata analysis
- Permission analysis
- Static application analysis
- AI malware detection
- Behavioral prediction

If a potential threat is detected, the system alerts the user and suggests actions such as:

- uninstalling the application
- restricting permissions
- User can Ignore The Alert and process to Application

The framework can also be integrated directly into Android devices, enabling smartphone manufacturers or enterprise systems to provide proactive security protection.

4. System Overview

BharatShield AI consists of two main components:

1. Security Framework

A background system that monitors application installations and performs security analysis.

2. Mobile Application

A user interface that displays scan results, threat alerts, and risk scores.

The system scans every newly installed application within **10–15 seconds** and determines whether it is safe or suspicious.

5. System Architecture

The architecture of BharatShield AI consists of multiple layers:

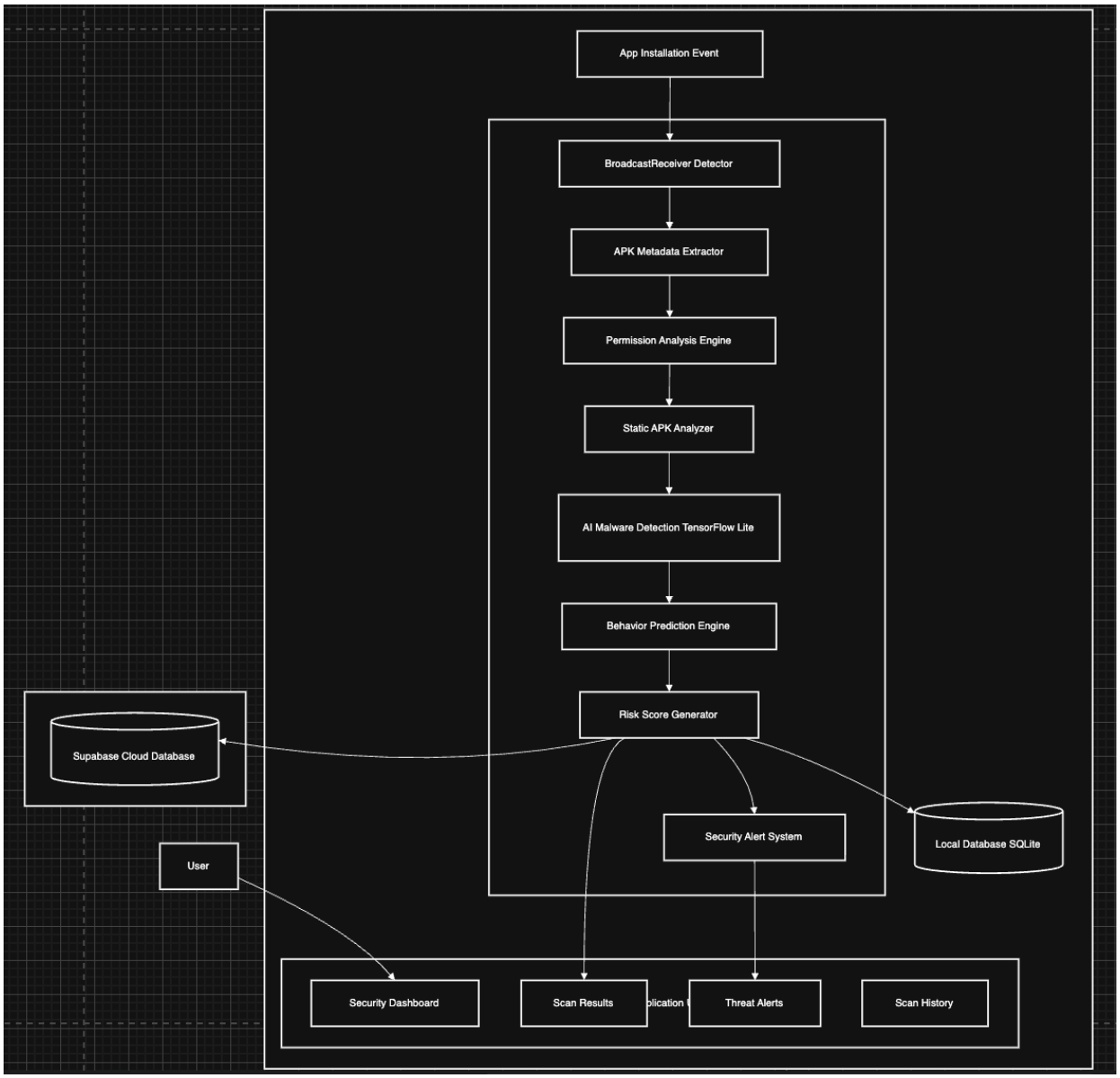

Figure 1: System Architecture of BharatShield AI

The above diagram represents the workflow of BharatShield AI, starting from application installation detection to risk score generation and user alert system. It also shows integration with local and cloud databases.

6. Key Innovations

1. Real-Time AI App Scanning

BharatShield scans applications immediately after installation and detects threats within **10–15 seconds**.

2. Explainable AI Security Analysis

The system explains why an application is dangerous instead of simply marking it as harmful.

Example:

Risk Score: 82%

Reasons:

Accessing SMS and Contacts

Hidden background activity

Unknown server communication

3. Behavioral Prediction Engine

The AI model predicts possible malicious behavior such as:

- OTP interception
- data exfiltration
- spyware activity

4. CrowdShield Collective Intelligence Network

Threat intelligence is shared anonymously between devices, allowing faster detection of malware.

5. Lightweight Mobile AI Framework

The AI detection model runs directly on the smartphone using TensorFlow Lite for fast and efficient analysis.

7. Technology Stack

Mobile Application

- React Native
- TypeScript
- Android Native Modules (for BroadcastReceiver)
- TensorFlow Lite (via Native Bridge)

Artificial Intelligence

- Python
- Scikit-learn
- TensorFlow
- • TensorFlow Lite (Mobile Integration via React Native Bridge)

APK Analysis

- Androguard
- APKTool
- Dex Parser

Cloud Backend

- Supabase
- PostgreSQL database

8. Framework and Application Design

Framework Layer

The BharatShield framework monitors Android system events.

When an application is installed:

1. Android triggers broadcast event

   ACTION_PACKAGE_ADDED

2. BroadcastReceiver detects installation.
3. APK metadata is extracted.
4. Application features are analyzed.
5. AI model predicts malware probability.

Application Layer

The BharatShield mobile application provides:

The mobile application is developed using React Native and TypeScript, enabling cross-platform UI with native Android integration for system-level event detection.

- Security dashboard
- Scan results
- Threat alerts
- Risk score explanation
- Scan history

9. Database Design

BharatShield uses a **hybrid database architecture**.

Local Database

Technology:

Room Database (SQLite)

Purpose:

Store local scan results.

Data stored:

- package name
- risk score
- scan result

- scan timestamp

Cloud Database

Technology:

Supabase (PostgreSQL)

Purpose:

Store global malware intelligence.

Tables include:

Malware Intelligence Table

Fields:

- app_hash
- package_name
- malware_type
- risk_score
- detected_features

Threat Reports Table

Stores suspicious app reports from devices.

Feature Pattern Table

Stores malicious feature patterns detected by AI.

10. Functional Requirements

The system must:

1. Detect new application installations
2. Extract APK metadata
3. Analyze application permissions
4. Perform static APK analysis

5. Generate machine learning feature vectors
6. Run AI malware detection model
7. Predict suspicious behavior
8. Generate risk score
9. Display explainable security alerts
10. Share threat intelligence through CrowdShield network

11. Non-Functional Requirements

Security

The system must not collect personal user data.

Reliability

The framework must continuously monitor installation events.

Usability

The interface must be simple for non-technical users.

Maintainability

The AI model should support future updates.

12. Interface Requirements

Frontend Framework:

- React Native (TypeScript)

User Interface

The application includes:

- Security dashboard
- Application scanner
- Threat alerts
- Risk score explanation

Software Interface

The system interacts with:

- Android OS
- TensorFlow Lite
- Supabase cloud database

13. Performance Requirements

Expected system performance:

Detection time: **10–15 seconds**

Detection accuracy: ~**95%**

Low CPU usage

Minimal battery consumption

14. Design Constraints

The system must operate within mobile device limitations such as:

- limited memory
- limited CPU resources
- Android OS security restrictions

15. Future Scope

Future improvements may include:

- dynamic behavioral monitoring
- federated AI learning
- global malware intelligence network
- enterprise mobile security platform
- cross-platform cybersecurity support

16. Conclusion

BharatShield AI is an intelligent mobile cybersecurity framework designed to detect malicious Android applications using Artificial Intelligence.

By combining **real-time scanning, explainable AI, behavioral prediction, and collaborative threat intelligence**, BharatShield AI provides proactive protection against modern mobile malware threats.

The system aims to enhance smartphone security while supporting the vision of **Atmanirbhar Bharat by developing indigenous cybersecurity technologies**.